NextHikes Advanced Scientific Calculator

PROJECT : NEXTHIKES IT SOLUTIONS

CREATED BY : MANJULA B

A GUI calculator with scientific functions,
calculation history, memory and keyboard support.



Project Overview

What the application does

A desktop scientific calculator built with Python and Tkinter.

Supports arithmetic, powers, percentages, trigonometry and logarithms.

Uses a custom expression evaluator rather than Python eval().

Includes a calculation-history window and memory controls.

Provides a themed interface and keyboard shortcuts.

Application Structure

Main modules and responsibilities

Window & UI

Tk root window, colors, display field, labels and button grid.

Math Engine

AST traversal, allowed operators, functions and constants.

Calculator Actions

Input handling, calculate, clear, backspace and percentage.

History

Stores expressions/results; lets users recall or clear entries.

Memory

MC, MR, M+ and M− operations for stored numeric values.

Keyboard Support

Maps keys to calculator actions using root.bind().

Safe Mathematical Engine

Expression parsing and validation

ast.parse(expression, mode='eval') converts the expression into a syntax tree.

The visitor accepts only numeric constants, approved operators, unary signs, named constants and approved function calls.

Operators are mapped through the operator module: +, −, ×, ÷, power and modulo.

Supported constants: π and e. Function examples: sin, cos, tan, inverse trig, sqrt, log, ln, abs, factorial and exp.

Guardrails reject unsupported syntax, oversized exponents, invalid factorial inputs, complex values and non-finite results.

Scientific & Core Operations

User-facing calculator capabilities

Basic arithmetic

Addition, subtraction, multiplication, division and modulo.

Scientific functions

sin/cos/tan, inverse trig, square root, powers, log and ln.

Additional actions

Percentage conversion, factorial, absolute value, exp, π and e.

Expression flow

Buttons append tokens to the expression string; calculate evaluates it.

Result formatting

Integers display without decimals; floats use compact precision.

Error handling

Invalid input shows “Error” and resets the active expression.

Calculation History & Memory

Convenience features

Calculation History

- Each successful calculation stores its expression and result.
- A separate window lists newest calculations first.
- Double-click a row to recall the result.
- Clear History removes the saved session list.

Memory Controls

- MC — clear stored memory.
- MR — recall memory into the display.
- M+ — add current expression value.
- M− — subtract current expression value.

Interface Design & Input

Tkinter components and interaction

Tkinter creates the fixed-size window, Entry display, Labels, Frames and Buttons.

A 4-column grid organizes the calculator keys into a compact layout.

Color palette: deep navy background, blue action keys, mint display and orange accent.

Keyboard binding supports digits and common operators; Enter calculates, Backspace deletes and Escape clears.

Button commands call the same action functions as keyboard events, keeping behavior consistent.



Conclusion & Future Enhancements

Summary

The project combines a user-friendly Tkinter interface with a guarded scientific expression engine.

History, memory and keyboard interactions improve usability.

The modular structure supports future maintenance and feature additions.

Possible enhancements: degree/radian toggle, better input validation, persistent history, themes, unit conversions and automated tests.

Thank you